

Machine Learning Fundamentals

Machine learning is a subfield of artificial intelligence that enables systems to learn patterns from data without being explicitly programmed. It is broadly divided into three categories: supervised learning, unsupervised learning, and reinforcement learning.

Supervised Learning

In supervised learning, the model is trained on labeled data, meaning each training example is paired with an output label. Common supervised learning algorithms include Linear Regression, Logistic Regression, Decision Trees, Random Forests, and Support Vector Machines. Supervised learning is typically divided into two types of problems: classification, where the output is a discrete category, and regression, where the output is a continuous value. For example, predicting whether an email is spam or not spam is a classification problem, while predicting house prices based on features like size and location is a regression problem.

Unsupervised Learning

Unsupervised learning deals with unlabeled data. The goal is to find hidden patterns or intrinsic structures in the input data. Common techniques include clustering algorithms like K-Means and Hierarchical Clustering, as well as dimensionality reduction techniques like Principal Component Analysis (PCA) and t-SNE. Clustering is often used in customer segmentation, anomaly detection, and organizing large sets of unlabeled documents.

Reinforcement Learning

Reinforcement learning involves an agent that learns to make decisions by taking actions in an environment to maximize cumulative reward. Key concepts include the agent, environment, state, action, and reward. Popular algorithms include Q-Learning, Deep Q-Networks (DQN), and Policy Gradient methods. Reinforcement learning has been famously used in game-playing AI, such as AlphaGo, and in robotics for teaching robots to walk or grasp objects.

Overfitting and Regularization

Overfitting occurs when a model learns the training data too well, including its noise, and performs poorly on new, unseen data. Techniques to prevent overfitting include cross-validation, regularization methods like L1 (Lasso) and L2 (Ridge) regularization, dropout in neural networks, and early stopping during training. Regularization adds a penalty term to the loss function to discourage overly complex models.